

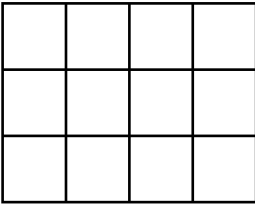
"Math is Cool" Masters -- 2025-26

6th Grade

Mental Math Solutions

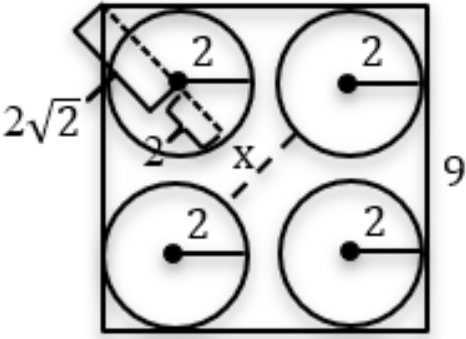
6th	Answer	Solution
1	50	What is twenty-five times four minus twenty-five times two? $25 \cdot 4 - 25 \cdot 2 = 50$
2	24	Evaluate ten X minus X squared if X equals four. $10 \cdot 4 - 4^2 = 24$
3	30 [inches]	The lengths of the two legs of a right triangle are five and twelve inches. What is the number of inches in the perimeter of the triangle? $c = \sqrt{5^2 + 12^2} = 13$, and $5 + 12 + 13 = 30$
4	31 [nickels]	Eho has one dollar and fifty-seven cents in his pocket. What is the largest number of nickels he could have in his pocket? $157/5 = 31 \text{ r}2$, so the answer is 31
5	3520 [feet]	There are five thousand two hundred and eighty feet in a mile. How many feet are in two-thirds of a mile? $5280/3 = 1760$ and $1760 \cdot 2 = 3520$
6	45	A sequence begins with the numbers three, six, ten, and fifteen. What is the eighth term in this sequence? Mentally list them out: 3, 6, 10, 15, 21, 28, 36, 45 Or use the formula $(n + 2)(n + 1)/2$, for $n = 8$ you get 45.
7	211	What is the largest three-digit prime number whose digits add up to four? Three-digit primes whose digits add up to 4 include: 103, 211 and 211 is the largest. FYI, 301 is divisible by 7.
8	[A + B =] 43	A standard deck of cards has all its face cards and aces removed. As a simplified fraction, the probability that two cards, randomly selected without replacement, are both red and divisible by three is A over B. What is A plus B? $P(\text{both divisible by } 3) = 6/36 \cdot 5/35 = 1/6 \cdot 1/7 = 1/42$, and $1 + 42 = 43$.

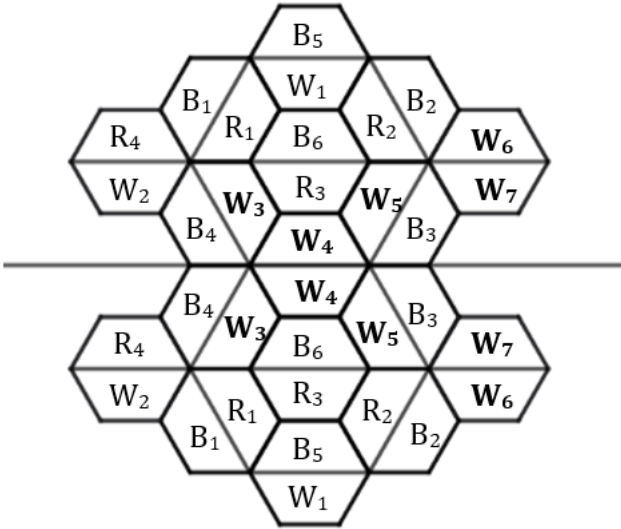
"Math is Cool" Masters -- 2025-26
6th Grade
Individual Test Solutions

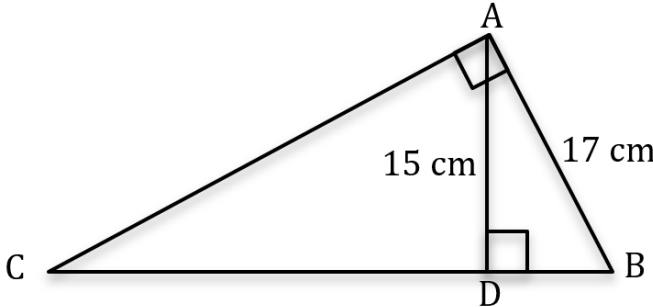
6th	Answer	Solution
1	56	What is 16% of 350? $16/100 \cdot 350 = 56$
2	144	Evaluate: $(\sqrt{144})^2$ $(\sqrt{144})^2 = 12^2 = 144$
3	99 [in ²]	How many square inches are in the area of a rectangle whose length is 11 inches and whose width is 9 inches? $9 \cdot 11 = 99$
4	[x =] 16	Solve for x: $7x + 61 = 173$ $7x + 61 = 173 \rightarrow 7x = 112 \rightarrow x = 16$
5	[A + B =] 23	As a simplified fraction, the ratio $820/123 = A/B$. What is A + B? $820/123 = 20/3$ and $20 + 3 = 23$
6	20 [squares]	How many squares of any size are in the figure? The figure is made up of unit squares combined to form a rectangle. $1 \times 1 - 12, 2 \times 2 - 6, 3 \times 3 - 2$ $12 + 6 + 2 = 20$ 
7	6 [= mean]	What is the mean of the following set of numbers? Your answer should be given as an integer. 5.4, 3.8, 9.6, 8.4, 2.8 $(5.4 + 3.8 + 9.6 + 8.4 + 2.8)/5 = 6$
8	168 [inches]	How many inches are in $4\frac{2}{3}$ yards? $4 \cdot 36 + \frac{2}{3} \cdot 36 = 144 + 24 = 168$
9	185 [minutes]	Ricardo reads 6 pages of his book every 5 minutes. At this rate, how long in minutes will it take him to complete his 222-page book? $222/6 = 37$ and $37 \cdot 5 = 185$
10	10 [%]	As a percentage, what is the probability that a number randomly selected from the integers 1 through 20 is a multiple of 7? The only multiples of 7 are 7 and 14, so the probability is $2/20 = 1/10$ and $1 + 10 = 11$.
11	42	Evaluate the expression $\frac{6a}{5} + \frac{12b}{7}$, if $a = 15$ and $b = 14$. $6 \cdot 15/5 + 12 \cdot 14/7 = 18 + 24 = 42$

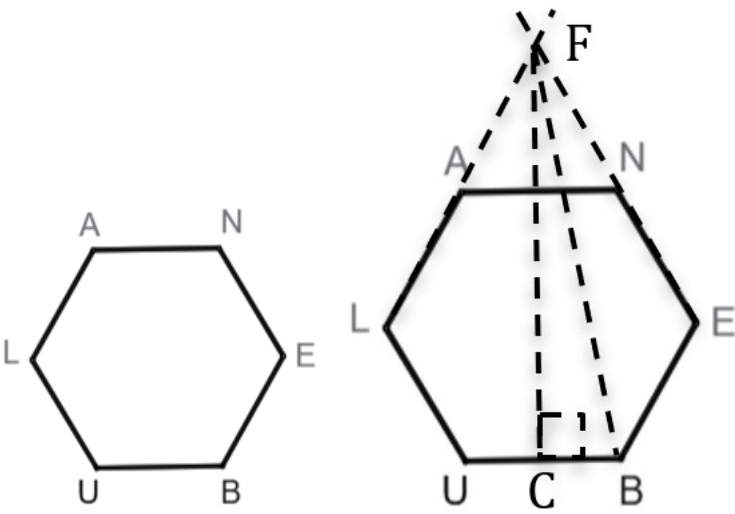
12	10235	What is the smallest positive 5-digit integer in which three of the digits are distinct odd integers and the other two digits are distinct even integers? The three smallest odd integers are 1, 3, and 5, and the smallest even integers are 0 and 2. Start with the smallest non-zero integer and then go in order from smallest to largest, which results in 10235.
13	98	What is the 10th term in an arithmetic sequence whose 2nd term is 18 and whose 5th term is 48? It goes up by 30 from the 2nd to the 5th term, so the common difference is $30/3 = 10$. The 10th term is $18 + 8 \cdot 10 = 98$
14	[CBA =] 981	In the following addition problem, each letter represents a digit from 0 to 9, and $A < B < C$. What is the largest possible three-digit integer CBA that can be formed? A + B = C Make C (9) as big as possible and make B (8) as big as possible. Then, $1 + 8 = 9$ results in a CBA value of 981.
15	9 [integers]	How many different four-digit integers can be formed by arranging the four digits in the number 2026? Note: a four-digit integer cannot have 0 in the thousands place. 2026, 2062, 2206, 2260, 2602, 2620, 6022, 6202, 6220 makes a total of 9
16	21000 [cm ²]	A rectangle has an area of 2.1 square meters. How many square centimeters are in the area of the rectangle? There are $100 \cdot 100 = 10000$ square centimeters in 1 square meter, so the answer is $2.1 \cdot 10000 = 21000$.
17	33	A data set consists of 7 distinct positive integers, has a range of 23 and a median of 13. What is the largest possible integer in the set? To make the largest number as large as possible, the four smallest numbers in the set would have to be 10, 11, 12, and 13. Then the largest number would be $10 + 23 = 33$.
18	126 [minutes]	It takes Nancy 3 hours to paint her room. It would take her little sister 7 hours to paint the same room. If they had worked together, in minutes, how long would it have taken them to paint the room? $1/3 + 1/7 = 1/x \rightarrow 7x + 3x = 21 \rightarrow 10x = 21 \rightarrow x = 2.1$ hours = 126 minutes
19	[a + b + c] = 30	The prime factorization of 460 can be written in the form $a^2 \cdot b \cdot c$, where a, b, and c are prime numbers. What is a + b + c? $460 = 2^2 \cdot 5 \cdot 23$, and $2 + 5 + 23 = 30$
20	6	Let A, B, C, and D, each represent a different positive even single-digit integer. What is the largest possible value of $\frac{A}{B} \cdot \frac{C}{D}$? Put the largest numbers in the numerators: $8/2 \cdot 6/4 = 6$

21	3 [= statement + no.]	Exactly one of the following statements is false, and the rest are true. Which statement is false? Answer 1, 2, 3, or 4. 1. Statement 2 is true. 2. Statement 3 is false. 3. Statement 4 is false. 4. Statement 1 is true. If statement 1 is false, then statement 2 is false, but there is only 1 false statement. If statement 2 is false, then statement 3 is true, which makes statement 4 false, but there is only 1 false statement. If statement 4 is false, then statement 1 is false, but there is only 1 false statement. If statement 3 is false, then statements 4, 1, and 3 are true. So, the answer is 3.
22	2	Evaluate: $\frac{\frac{2}{3}}{\frac{5}{7}} \cdot \frac{1}{6} + \frac{83}{45}$ $\frac{\frac{2}{3}}{\frac{5}{7}} \cdot \frac{1}{6} + \frac{83}{45} = \frac{14}{15} \cdot \frac{1}{6} + \frac{83}{45} = \frac{7}{45} + \frac{83}{45} = 90/45 = 2$
23	-35	If $a \Omega b = (3 - a)(5 - b)$, then evaluate $10 \Omega (20 \Omega 5)$. $(3 - 20)(5 - 5) = 0$ and $(3 - 10)(5 - 0) = -35$
24	[x =] 90 [%]	Blake bought a car for \$30000. Later, he sold it for 20% more than he paid for it. Using all the money he got from selling the car, he bought a new, more expensive car. He then sold the second car for an amount, x, that is 25% less than he paid for it. What percentage of the original \$30000 purchase price is x? Sold the first car for $1.2 \cdot 30000 = \\$36000$. Sold the 2nd car for $0.75 \cdot 36000 = \\$27000$. $27000/30000 = 0.9 = 90\%$.
25	5 [miles]	Maria walks $\frac{1}{3}$ of a mile in $\frac{1}{7}$ of an hour. At this average rate, how far in miles would she walk in $2\frac{1}{7}$ hours? $\frac{1}{3}$ mi in $\frac{1}{7}$ h $\rightarrow \frac{7}{3}$ mi in 1 h $\rightarrow 105/21 = 5$ mi in $15/7$ h

26	20 [ways]	How many ways are there to arrange the letters in the word ALIEN, if the vowels must always be in alphabetical order (not necessarily consecutive) from left to right? The three vowels can be all together 3 ways: AEI _ _ _ AEI _ _ _ AEI The three vowels can be a single and a pair that are together in 6 ways: A _ EI _ A _ _ EI _ A _ EI AE _ I _ AE _ _ I _ AE _ I The three vowels can be three separate single letters in 1 way: A _ E _ I Each of these 10 arrangements can have the L and the N in two arrangements, so the final answer is $10 \cdot 2 = 20$.
27	14 [= 100 th term]	The first 6 terms of an infinite sequence are 1, 2, 2, 3, 3, 3, . . . What is the 100th term in this sequence? If n represents the value of the largest term in the sequence, then $n(n + 1)/2 = 91$ total terms when $n = 13$. Therefore, the next 14 terms will be 14, so the 100th term is 14.
28	4 [circles]	A square has an area of 81 cm^2. How many circles of radius 2 cm can fit completely inside the square without overlapping? The diagonal of the square is $9\sqrt{2}$ and let x = the length of the diagonal space between two circles represented by the dashed line in the figure. Then $x = 9\sqrt{2} - [2(2\sqrt{2} + 2)] = 9\sqrt{2} - (4\sqrt{2} + 4) = 5\sqrt{2} - 4 \approx 5(1.4) - 4 \approx 7 - 4 \approx 3$, which is less than 4 so only 4 circles will fit. 
29	[A + B =] 598	All face cards and aces are removed from a standard deck of cards. As a simplified fraction, the probability that a hand of three cards drawn from the remaining cards consists of three of a kind is A/B. What is $A + B$? Note: three of a kind means something like three 2s or three 10s. $36/36 \cdot 3/35 \cdot 2/34 = 3/595$ and $3 + 595 = 598$

30	[n =] 49	What is the smallest positive integer n, for which each of the expressions $\frac{n+2}{3}$, $\frac{3n+1}{4}$, and $\frac{2n+7}{5}$ represents an integer? Values of n that make $\frac{2n+7}{5}$ an integer: 4, 9, 14, 19, 24, 29, ... In this list every number ends with a 4 or a 9. Find the first number that the next two lists have in common that ends in a 4 or a 9. Values of n that make $\frac{n+2}{3}$ an integer and that end in 4 or 9: 4, 19, 34, 49 Values of n that make $\frac{3n+1}{4}$ an integer and that end in 4 or 9: 9, 29, 49 The answer is 49.
31	[A + B =] 17	Athena, Joy, and Mila simultaneously begin jogging in the same direction from the same point on a circular track. Joy's average speed is 10% slower than Athena's average speed. Athena passes Mila for the first time at the moment when Athena completes her fifth lap. Joy's speed is $A.B\%$ faster than Mila's speed, where A is a 2-digit integer and B is a 1-digit integer. What is $A + B$? When Athena passes Mila, Athena has run 5 laps, and Mila has run 4 laps. Since Joy runs 10% slower than Athena, she has completed 4.5 laps at that moment. Then Joy's speed is $0.5/4 = 1/8 = 12.5\%$ faster than Mila's speed, and $12 + 5 = 17$.
32	5 [white pairs]	Each half of this figure is composed of 4 red trapezoids, 6 blue trapezoids and 7 white trapezoids. When the upper half is folded down over the centerline, 2 pairs of red trapezoids coincide, as do 4 pairs of blue trapezoids. There are 2 blue-white pairs. How many white pairs coincide? There are 4 coinciding pairs of blue trapezoids, labeled in the diagram below B_1, B_2, B_3, and B_4, in both the top and the bottom half. The 2 blue-white pairs have a blue from the top half coinciding with a white from the bottom half, labeled B_5 and W_1, and a white from the top half coinciding with a blue from the bottom half labeled W_1 and B_5. This makes a total of 5 blue trapezoids in each half. Therefore, there must be two blue-red pairs, a blue from the top half coinciding with a red from the bottom half, labeled B_6 and R_3, and a red from the top half coinciding with a blue from the bottom half labeled R_3 and B_6. This makes a total of 3 reds in each half, so there must still be 2 red-white pairs, a red from the top half coinciding with a white from the bottom half, labeled R_4 and W_2, and a white from the top half coinciding with a red from the bottom half labeled W_2 and R_4. This leaves room for 5 pairs of coinciding white trapezoids, and these are labeled with bolded W_3, W_4, W_5, W_6, and W_7, in both the top and the bottom half. So, the final answer is 5. 

33	[A + B =] 89	A jar contains 4 red, 3 blue, and 2 green marbles. Three marbles are randomly removed from the jar. As a simplified fraction, the probability that all three marbles are of the same color is A/B. What is $A + B$? $P(RRR) = 4/9 \cdot 3/8 \cdot 2/7 = 1/21$ $P(BBB) = 3/9 \cdot 2/8 \cdot 1/7 = 1/84$ $P(GGG) = 0$ $1/21 + 1/84 = 4/84 + 1/84 = 5/84$ and $5 + 84 = 89$
34	[P + Q =] 319	In $\triangle ABC$, $\overline{AD} \perp \overline{BC}$, $m\angle BAC = 90^\circ$, $AB = 17$ cm, and $AD = 15$ cm. As a simplified fraction, the ratio of $BD/AC = P/Q$. What is $P + Q$? $BD = \sqrt{17^2 - 15^2} = 8$ and since the triangles are similar, $8/17 = 15/AC \rightarrow AC = 255/8$ and $BD:AC = 8:255/8 = 64/255$ and $64 + 255 = 319$. 
35	[Q =] 196	On a coordinate plane, $\triangle ABC$ has vertices $A(-4, 0)$, $B(1, 7)$, and $C(8, 0)$. Imagine this triangle is rotated 360° about the x-axis. The volume of the 3-dimensional space through which the triangle rotates can be expressed as $Q\pi$ cubic units. What is the value of Q? The 3-dimensional space through which the triangle rotates consists of two cones which share a base. The radius of the base is 7, the height of one cone is 5 and the height of the other cone is 7. The volume of this space is $7^2\pi \cdot 5/3 + 7^2\pi \cdot 7/3 = 245\pi/3 + 343\pi/3 = 588\pi/3 = 196\pi$, and $Q = 196$.
36	162 [integers]	How many positive 4-digit integers have no repeating digits, no digit that is a 2, and have digits whose sum is 24? The 9 possible sets of 4 distinct digits that add to 24 include: 9870, 9861, 9852, 9843, 9762, 9753, 9654, 8763, and 8754. Seven of the sets do not have a 2 and each set has $4! = 24$ arrangements, except for the set that has a 0, in which case it has 18 arrangements, because the 6 arrangements in which it has a 0 in the thousands place would not be 4-digit integers. So, the final answer is $6 \cdot 24 + 18 = 162$.
37	2	A number, n, is 2 more than 3 times its reciprocal. What is the sum of all values of n? Let n be the number that is 2 more than 3 times its reciprocal. Then $n = 3/n + 2 \rightarrow n^2 = 3 + 2n \rightarrow n^2 - 2n - 3 = 0 \rightarrow (n - 3)(n + 1) = 0$, so $n = 3$ and $n = -1$, and $3 + -1 = 2$
38	[n =] 42	What is the maximum value of n, such that 8^n is a factor of $130!$ (130 factorial)? Determine how many total factors of 2 there are in $130!$ and then divide by 3, since 3 factors of 2 is the same as 1 factor of 8. There are 65 multiples of 2, which means there are 65 factors of 2. There are 32 multiples of 4, which increases the number of factors of 2 by 32. There are 16 multiples of 8, which increases the number of factors of 2 by 16. There are 8 multiples of 16, which increases the number of factors of 2 by 8. There are 4 multiples of 32, which increases the number of factors of 2 by 4. There are 2 multiples of 64, which increases the number of factors of 2 by 2. There is 1 multiple of 128, which increases the number of factors of 2 by 1. The total number of factors of 2 is $1 + 2 + 4 + 8 + 16 + 32 + 65 = 128$, so the total number of factors of 8 is $128/3$, which is 42 r2. Therefore, the answer is 42.

39	1983 [= median]	What is the median of the following list of 2026 numbers? $2, 4, 6, \dots, 2026, 2^2, 4^2, 6^2, \dots, 2026^2$ $44^2 = 1936$ and $46^2 > 2026$. This means that 2026 is the $(1013 + 22 =) 1035^{\text{th}}$ term in the list. To get the median we need the 1013^{th} and 1014^{th} terms. Since the non-square terms increase by 2, we subtract $22 \cdot 2 = 44$ and $21 \cdot 2 = 42$ from 2026 to get 1982 and 1984 as the 1013^{th} and 1014^{th} terms. Therefore, the median is $(1982 + 1984)/2 = 1983$.
40	$[R + S + T =] 8$	The lengths of the three sides of a right triangle form a geometric sequence. The ratio of the length of the hypotenuse to the length of the short leg can be expressed as $\frac{R+\sqrt{S}}{T}$, where R, S, and T are each single-digit integers. What is $R + S + T$? If the 3 sides of the triangle are a, b, and c, then $a^2 + b^2 = c^2$, and if a, b, and c form a geometric sequence, then $c = a \cdot r^2$ and $b = a \cdot r$, where r is the common ratio of the geometric sequence. Substituting for c and b, the resulting equation is $a^2 + a^2r^2 = a^2r^4 \rightarrow 1 + r^2 = r^4 \rightarrow r^4 - r^2 - 1 = 0$. Let $p = r^2$, then $p^2 - p - 1 = 0$, and solving using the quadratic formula results in $p = \frac{1+\sqrt{5}}{2} = r^2$, and since $c = a \cdot r^2$, then the ratio of the hypotenuse to the short leg, $c/a = r^2$. So, the answer is $1 + 5 + 2 = 8$. Note: p cannot equal $\frac{1-\sqrt{5}}{2}$, because $p = r^2$, and r^2 cannot be negative.
41	$[P + Q =] 19$	Sides $\overline{LA}$ and $\overline{EN}$ in a regular hexagon <i>NEBULA</i>, shown here, are extended until they intersect at a point F. If the sides of the hexagon have length 12 units, then the length of $\overline{FB}$ is $P\sqrt{Q}$ units, where P and Q are both integers, and Q has no factors that are perfect squares, other than 1. What is $P + Q$? Add in the lines to form right $\triangle CBF$. To determine CF, notice $\triangle AFN$ is an equilateral triangle with sides of length 12, so its height is $6\sqrt{3}$. Then, because a regular hexagon can be divided into 6 congruent equilateral triangles that are congruent to $\triangle AFN$, then $CF = 3 \cdot 6\sqrt{3} = 18\sqrt{3}$ and $CB = 6$. So, $FB = \sqrt{(18\sqrt{3})^2 + 6^2} = \sqrt{324 \cdot 3 + 36} = \sqrt{972 + 36} = \sqrt{1008} = 12\sqrt{7}$, so the answer is $12 + 7 = 19$. 

$[a_3 - b_3 =]$
12

Two distinct geometric sequences consist of three integers. One is a_1, a_2, a_3 , and the other is b_1, b_2, b_3 , both sets of three integers have a mean of 21, and $a_1 < b_1$.

What is the value of $a_3 - b_3$?

Let $x = a_1$ or b_1 and $r =$ the common ratio. Then you can write the equation $x + xr + xr^2 = 63 \rightarrow r^2 + r + 1 = 63/x$.

Substitute values of x and solve for r . Any values of x that are not factors of 63, will result in an r -value that is not an integer, and therefore will lead to xr and/or xr^2 values that are not integers. So, we only have to substitute factors of 63 for x .

Case 1: $x = 1 \rightarrow r^2 + r + 1 = 63 \rightarrow r^2 + r - 62 = 0 \rightarrow r = \frac{-1 \pm \sqrt{1+248}}{2}$, which is an irrational number, so ar will not be an integer.

Case 2: $x = 3 \rightarrow r^2 + r + 1 = 21 \rightarrow r^2 + r - 20 = 0 \rightarrow (r + 5)(r - 4) = 0 \rightarrow r = -5$ and $r = 4$. The common ratio needs to be positive, so if $x = 3$ and $r = 4$, resulting in a series 3, 12, 48 where the sum is 63.

Case 3: $x = 7 \rightarrow r^2 + r + 1 = 9 \rightarrow r^2 + r - 8 = 0 \rightarrow r = \frac{-1 \pm \sqrt{1+32}}{2}$, which is an irrational number, so ar will not be an integer.

Case 4: $x = 9 \rightarrow r^2 + r + 1 = 7 \rightarrow r^2 + r - 6 = 0 \rightarrow (r + 3)(r - 2) = 0 \rightarrow r = -3$ and $r = 2$. The common ratio needs to be positive, so if $x = 9$ and $r = 2$, resulting in a series 9, 18, 36 where the sum is 63.

Case 5: $x = 21 \rightarrow r^2 + r + 1 = 3 \rightarrow r^2 + r - 2 = 0 \rightarrow (r + 2)(r - 1) = 0 \rightarrow r = -2$ and $r = 1$. When $r = 1$ there is no geometric sequence.

Case 6: $x = 63 \rightarrow r^2 + r + 1 = 1 \rightarrow r^2 + r = 0 \rightarrow r(r + 1) = 0 \rightarrow r = 0$ and $r = -1$. Neither of these will produce a geometric sequence.

Something else to consider as a possible x -value would be powers, like perfect square, perfect cubes, etc., because then r can be a rational number with a denominator that is the square root of the perfect square, or cube root of the perfect cube, etc.

Perfect squares:

If $x = 4$, r can be $3/2, 5/2, 7/2$, etc. Possible sequences include: if $r = 5/2$, 4, 10, 25, sum = 39, or if $r = 7/2$, 4, 14, 49, sum = 67. Since one has a sum that is greater and one has a sum that is lesser, x cannot be 4.

We do not need to check 9, since one of the sequences we have begins with 9.

If $x = 16$ and $r = 5/4$, the result is 16, 20, 25, and the sum is 61, if $r = 7/4$, the result is 16, 28, 49, and the sum is 93, so x cannot be 16.

If $x = 25$, and $r = 6/5$, the result is 25, 30, 36, which has a sum that is greater than 63, so we don't need to check any other perfect squares, since they will all lead to sums greater than 63.

Perfect cubes:

If $x = 8$ and $r = 3/2$, the result is 8, 12, 18 and the sum is 38, if $r = 5/2$, the result is 8, 20, 50, and the sum is 78, so x cannot be 8.

If $x = 27$ and $r = 4/3$, the result is 27, 36, 48, and the sum is greater than 63, so x cannot be 27.

Perfect fourths:

The smallest perfect fourth is 16 and we already checked it as a perfect square.

The next perfect fourth is 81, which will lead to a sum that is greater than 63.

Perfect fifths will also lead to sums greater than 63, so we can conclude that there are just the 2 sequences.

So, a_1, a_2, a_3 , corresponds with 3, 12, 48, and b_1, b_2, b_3 , corresponds with 9, 18, 36, since $a_1 < b_1$. Therefore, the final answer will be $a_3 - b_3 = 48 - 36 = 12$.

43

101

What is the sum of all prime numbers less than 40 that are the largest value in a Pythagorean Triple? An example is 5, since it is prime and is the largest value in the 3:4:5 Pythagorean Triple.

You can generate Pythagorean Triples by choosing two positive integer values, m and n , such that $m > n$, $GCF(m, n) = 1$, and either m or n is even, but not both. Then, the Pythagorean Triple consists of $m^2 - n^2$, $2mn$, and $m^2 + n^2$. Here is a table to keep track:

m	n	$m^2 - n^2$	$2mn$	$m^2 + n^2$
2	1	3	4	5
3	2	5	12	13
4	1	15	8	17
4	3	7	24	25
5	2	21	20	29
5	4	9	40	41
6	1	35	12	37
6	3	27	36	45
6	5	11	60	61
7	2	45	28	53
7	4	33	56	65
7	6	13	84	85

Continuing the table results only in hypotenuse values that are larger than 40. So, the answer is $5 + 13 + 17 + 29 + 37 = 101$

44

$[r + s + t =]$

9

The solutions $x = u$ and $x = v$ of the quadratic equation $rx^2 + sx + t = 0$ are reciprocals of the solutions of the quadratic equation $(2 + a)x^2 + 5x + (2 - a) = 0$ for some integer a . If the GCF of r , s and t is 1, what is the value of $r + s + t$?

The quadratic equations $ax^2 + bx + c = 0$ and $cx^2 + bx + a = 0$ have solutions that are reciprocals of each other, in other words when a and c swap places. So, if the solutions of $rx^2 + sx + t = 0$ are the reciprocals of the solutions of $(2 + a)x^2 + 5x + (2 - a) = 0$, then they are equal to the solutions of $(2 - a)x^2 + 5x + (2 + a) = 0$ (the coefficient of x^2 and the constant swapped places). Therefore, $r = 2 - a$, $s = 5$, and $t = 2 + a$. Plugging in integers for a always results in a sum of $r + s + t = 9$. See table:

a	$r = 2 - a$	$t = 2 + a$	s	$r + s + t$
...	...	...	5	9
-3	5	-1	5	9
-2	4	0	5	9
-1	3	1	5	9
0	2	2	5	9
1	1	3	5	9
2	0	4	5	9
3	-1	5	5	9
...	...	...	5	9

So, the answer is 9.

45	[n =] 19	For years, Mr. Kangas has had a tradition of giving his students a list of books from which they had to choose two to read during spring break. This year, Mr. Kangas added n book(s) to the list. If there are now 84 more ways that two books can be selected, what is the sum of all possible values of n? The number of ways to choose 2 books from a list of n books is ${}_nC_2$. The difference between ${}_nC_2$ and ${}_{n+1}C_2$ is n, the difference between ${}_{n+1}C_2$ and ${}_{n+2}C_2$ is $n+1$, and so on. So, you're looking for how many different sets of consecutive integers add up to 84. Since there must be at least 2 books to start with, the largest list of consecutive numbers that could possibly add up to 84 would be 11 numbers, because the sum of the 11 numbers from 2 to 12 = 77 and the sum of the 11 numbers 3 to 13 = 88. Neither of these is 84 so n cannot be 11. The value of n cannot equal 10 because $3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 = 75 \neq 84$ and $4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 = 85 \neq 84$. The value of n cannot equal 9 because $5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 = 81 \neq 84$ and $6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 = 90 \neq 84$. The value of n can equal 8 because $7 + 8 + 9 + 10 + 11 + 12 + 13 + 14 = 84$, when the number of books increases from 7 to 15, the calculation is ${}_{15}C_2 - {}_7C_2 = 7 \cdot 15 - 7 \cdot 3 = 7 \cdot 12 = 84$. The value of n can equal 7 because $9 + 10 + 11 + 12 + 13 + 14 + 15 = 84$, when the number of books increases from 9 to 16, the calculation is ${}_{16}C_2 - {}_9C_2 = 8 \cdot 15 - 9 \cdot 4 = 120 - 36 = 84$. The value of n cannot equal 6 because $11 + 12 + 13 + 14 + 15 + 16 = 81 \neq 84$ and $12 + 13 + 14 + 15 + 16 + 17 = 87 \neq 84$. The value of n cannot equal 5 because $14 + 15 + 16 + 17 + 18 = 80 \neq 84$ and $15 + 16 + 17 + 18 + 19 = 85 \neq 84$. The value of n cannot equal 4 because $19 + 20 + 21 + 22 = 82 \neq 84$ and $20 + 21 + 22 + 23 = 86 \neq 84$. The value of n can equal 3 because $27 + 28 + 29 = 84$, when the number of books increases from 27 to 30, the calculation is ${}_{30}C_2 - {}_{27}C_2 = 29 \cdot 15 - 27 \cdot 13 = 435 - 351 = 84$. The value of n cannot equal 2 because $41 + 42 = 83 \neq 84$ and $42 + 43 = 85 \neq 84$. The value of n can equal 1 when the number of books increases from 84 to 85, the calculation is ${}_{85}C_2 - {}_{84}C_2 = 85 \cdot 42 - 83 \cdot 42 = 2 \cdot 42 = 84$ So, the answer is $1 + 3 + 7 + 8 = 19$.
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“Math is Cool” Masters -- 2025-26

6th Grade

Multiple Choice Solutions

6th	Answer	Solution
USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #4. If $a \Delta b = 3ab$, then $2 \Delta 4 = 3 \cdot 2 \cdot 4 = 24$		

1	C	If $6 \square 9 = 45$, then which of the following rules could define $a \square b$? A) $a \square b = 5a + 2b$ B) $a \square b = 10a - 2b$ C) $a \square b = b^2 - a^2$ D) $a \square b = a^2 + b^2 - 70$ E) Answer not given. Since $9^2 - 6^2 = 45$, the correct definition is $b^2 - a^2$
2	A	If $a \odot b = a(a + b)/2$, then what is $5 \odot 7$? A) 30 B) 40 C) 50 D) 60 E) Answer not given. $5(5 + 7)/2 = 30$
3	C	Using the functions as defined above, what is the value of: $8 \Delta (6 \square (4 \odot 2))$? A) 864 B) 1744 C) 2592 D) 12960 E) Answer not given. $4(4 + 2)/2 = 12$ and $12^2 - 6^2 = 108$ and $3 \cdot 8 \cdot 108 = 2592$

4

D

Let a be the arithmetic sequence $a_1, a_2, a_3, \dots, a_n, \dots = 3, 6, 9, \dots$ and let b be the arithmetic sequence $b_1, b_2, b_3, \dots, b_n, \dots = 15, 13, 11, \dots$ and let $f(x) = a_x \square b_x = 8a_x - 5b_x$.

For example, $f(1) = a_1 \square b_1 = 8a_1 - 5b_1 = 8 \cdot 3 - 5 \cdot 15 = 24 - 75 = -51$.

Which of the following is $f(x)$ in terms of x ?

A) $f(x) = -1.5x^2 + 38.5x - 88$ B) $f(x) = -0.5x^2 + 35.5x - 86$

C) $f(x) = 33x - 84$

D) $f(x) = 34x - 85$

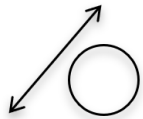
E) Answer not given.

From the table below, the slope is $(17 - (-17))/(3 - 2) = 34$ and $f(0) = -51 - 34 = -85$, so the function is $f(x) = 34x - 85$.

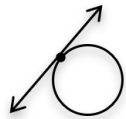
x	$a_x \square b_x$	$f(x)$
1	$8 \cdot 3 - 5 \cdot 15$	-51
2	$8 \cdot 6 - 5 \cdot 13$	-17
3	$8 \cdot 9 - 5 \cdot 11$	17

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #5 THROUGH #7.

The first set of figures shown demonstrate the different number of points of intersection possible for a line and a circle.



0 points of intersection

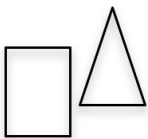


1 point of intersection



2 points of intersection

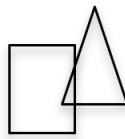
The next set of figures shown demonstrate the different number of points of intersection possible for two or more shapes with straight edges. Note: in the 4th and 5th figures below, one side of the triangle lies directly on top or partly on top of one of the sides of the rectangle.



0 points of intersection



1 point of intersection



2 points of intersection



infinite points of intersection



infinite points of intersection

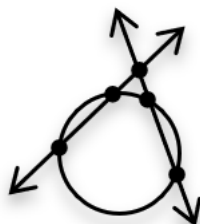
5

E (5)

What is the maximum possible number of points of intersection for two lines and a circle? Note: points of intersection can be where one line and the circle intersect, where the two lines intersect each other, or where both lines and the circle intersect. Also, the lines do not lie on top of each other.

A) 1 B) 2 C) 3 D) 4 E) Answer not given.

5 points of intersection are possible as shown here.

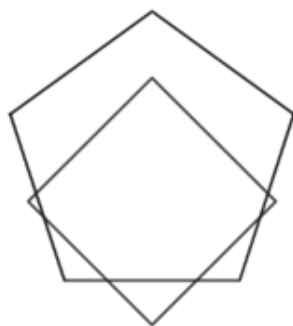


6**C**

What is the maximum possible number of points of intersection for a square and a regular pentagon whose sides are the same length as the sides of the square? Note: no side of the square lies on top of part or all of a side of the pentagon.

A) 4 B) 5 C) 6 D) 7 E) Answer not given.

The answer is 6. See figure below.



7

D

Imagine all the ways you can draw a line, a square, and a circle of any size, either overlapping or not overlapping. Let N equal the number of points of intersection that can be created by drawing these three shapes. What is the sum of all possible values of N ? Note: the line does not lie on top of any of the sides of the square.

A) 57

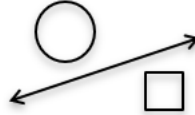
B) 60

C) 67

D) 78

E) Answer not given.

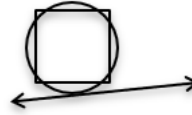
0 points:



5 points:



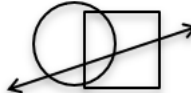
9 points:



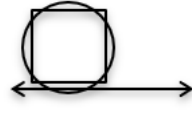
1 point:



6 points:



10 points:



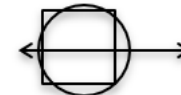
2 points:



7 points:



11 points:



3 points:



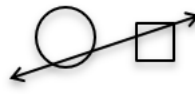
8 points:



12 points:



4 points:



12 is the maximum

So, N can be anything from 0 to 12, and $0 + 1 + 2 + \dots + 11 + 12 = 78$.

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #8 THROUGH #10.

A popular math game is called "ZIP, ZAP, ZORP" - maybe you have played it before! In the game, one person chooses a 2-digit whole number with no repeated digits, and another person tries to guess the number. After each guess, the person who chose the number will give one of the following clues:

- ZIP: No digit is correct
- ZAP: One digit is correct, and it is in the right place
- ZORP: One digit is correct, but it is in the wrong place.
- ZORP ZORP: Two digits are correct, but they are both in the wrong place.
- ZAP ZAP: Correct number!

Examples: Player A chooses 79 as the secret number. Player B guesses 42. Player A says 'ZIP'.
OR if Player B guesses 91, Player A says 'ZORP'.

8	B	Biff picks 43 as his secret number. Eho's first guess is 40. What is Biff's response? A) ZIP B) ZAP C) ZORP D) ZORP ZORP E) Answer not given. One digit is correct (3), and it is in the right place.
9	B	Based on the events in problem 8, how many reasonable 2nd guesses are available to Eho? Note: a reasonable 2nd guess is one that uses the information gained from the response to the 1st guess to narrow down the list of choices as much as possible. A) 15 B) 16 C) 17 D) 18 E) Answer not given. The knowledge that comes from the response to the first guess (ZAP) is that one of the digits, 4 or 0, is correct. If 4 is correct, then 0 is not, so the reasonable guesses having a 4 in the tens place include 41, 42, 43, 45, 46, 47, 48, and 49. If 0 is correct, then 4 is not, so the reasonable guesses having a 0 in the ones place include 10, 20, 30, 50, 60, 70, 80, and 90. There are a total of $8 + 8 = 16$ reasonable 2nd guesses.

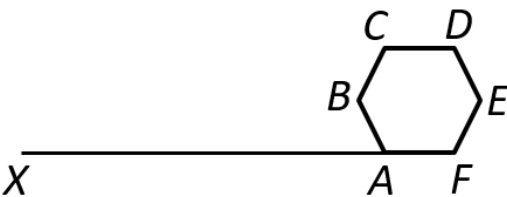
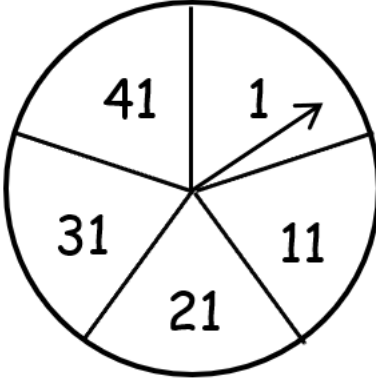
10	D	Continuing the game described in problems 8 and 9, Eho makes a reasonable 2nd guess, to which Biff responds with ZIP. Assuming Eho uses perfect logic, what is the probability that he guesses the right number on his 3rd guess? A) 1/10 B) 1/9 C) 1/8 D) 1/7 E) Answer not given. Reasonable 2nd guesses that would result in a ZIP response would be 10, 20, 50, 60, 70, 80, and 90. From the response of ZIP, Eho can infer that the tens digit of the correct number is a 4. If his 2nd guess was any of these numbers that resulted in a ZIP response, then he can eliminate one from the list 41, 42, 43, 45, 46, 47, 48, 49, because of the tens digit from his 2nd guess, so the probability of getting the correct number on his 3rd guess would be 1/7.
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“Math is Cool” Masters -- 2025-26

6th Grade

Team Test Solutions

6th	Answer	Solution
1	54	What is the sum of all positive prime numbers less than 50 with at least one digit that is a 2? $2 + 23 + 29 = 54$
2	80	Evaluate: $10000 \cdot (0.2)^3$ $10000 \cdot (0.2)^3 = 80$
3	$[x + y =]$ 88	Find the sum $x + y$ of the numbers in the following sequence: 71, 62, x , 44, y , 26, 17, ... The numbers are decreasing by 9, to the sequence is 71, 62, 53, 44, 35, 26, 17, ..., and $x + y = 53 + 35 = 88$
4	82	A class of eight students had an average test score of 86. One student scored a perfect score of 100 and another received a 96. What was the mean test score of the other 6 students? $8 \cdot 86 = 688$ and $688 - 100 - 96 = 492$ and $492/6 = 82$
5	2 [miles]	A tortoise walks at an average rate of 2 miles every 3 hours. A hare walks at an average rate of $2/3$ mile every 15 minutes. At these rates and in miles, how much farther will the hare walk in an hour than the tortoise? The hare walks $2/3$ mile in 15 min = $8/3$ mile in an hour The tortoise walks 2 miles in 3 hours = $2/3$ mile in an hour $8/3 - 2/3 = 6/3 = 2$.
6	$31_{[10]}$	Evaluate $456_7 - 345_6 - 234_5$ as a base-10 number. $4 \cdot 49 + 5 \cdot 7 + 6 \cdot 1 - (3 \cdot 36 + 4 \cdot 6 + 5 \cdot 1) - (2 \cdot 25 + 3 \cdot 5 + 4 \cdot 1) =$ $196 + 41 - (108 + 29) - (50 + 19) = 237 - 137 - 69 = 31$

7	$[R + T =]$ 185	A twelve-foot-long string, $\overline{AX}$, is attached to vertex A of regular hexagon $ABCDEF$, which has a perimeter of 12 feet. The string is kept taut (pulled straight), and wrapped around the hexagon, by moving in a circular clockwise direction. The area swept out by the string as it moves around the hexagon can be written as $R\pi/T$, where R and T are positive integers whose greatest common factor is 1. What is $R + T$? The first rotation will be 60°, or $1/6$ of a circle, and the radius will be 12, the next rotation will also be 60°, and the radius will be 10, and so on. The calculation will be $12^2\pi/6 + 10^2\pi/6 + 8^2\pi/6 + 6^2\pi/6 + 4^2\pi/6 + 2^2\pi/6 = (144 + 100 + 64 + 36 + 16 + 4)\pi/6 = 364\pi/6 = 182\pi/3$, and $182 + 3 = 185$. 
8	$[A + B =]$ 34	A circular spinner has 5 congruent sections, each labeled with one of the numbers, 1, 11, 21, 31, and 41. The spinner is spun twice, and the resulting numbers are recorded. As a simplified fraction, the probability that the sum of the two numbers is divisible by 3 is A/B. What is $A + B$? There are $5 \cdot 5 = 25$ possible pairs of numbers. Pairs whose sum is divisible by 3 include: (1, 11), (11, 1), (1, 41), (41, 1), (11, 31), (31, 11), (21, 21), (31, 41), and (41, 31), so the answer is $9/25$ and $9 + 25 = 34$. 
9	$[A + B =]$ 41	An equilateral triangle and a regular hexagon have a combined perimeter of 54 cm and a combined area of $81\sqrt{3}$ cm². As a simplified fraction, the side length of the hexagon is A/B cm. What is $A + B$? Let a = the side length of the triangle and b = the side length of the hexagon. Then $3a + 6b = 54 \rightarrow a = 18 - 2b$, and $\frac{a^2\sqrt{3}}{4} + \frac{6b^2\sqrt{3}}{4} = 81\sqrt{3} \rightarrow a^2 + 6b^2 = 324$. Combining the two equations results in $324 - 72b + 4b^2 + 6b^2 = 324 \rightarrow 10b^2 - 72b = 0 \rightarrow 2b(5b - 36) = 0 \rightarrow b = 0$ and $b = 36/5$. To be the length of a side of the hexagon, b cannot equal 0, so $b = 36/5$ and $36 + 5 = 41$.

10

27

From a standard deck of playing cards, all 13 hearts have been lined up in a row, with positions 1 through 13, as indicated below. Face cards are defined as Jack, Queen and King. The Ace is the high card, so the natural order of the cards from left to right is 2 through 10, Jack, Queen, King, and Ace. Assume a value of 15 for the ace and face cards. Otherwise, assume their numerical value.

Given the following clues, determine the order of the 13 cards.

What is the sum of the card values in positions 5, 7 and 8?

Clues:

- No two cards are in their natural order from left to right.
- No face cards are at either end.
- Two face cards are adjacent.
- The King and Ace are separated by one card.
- The 2nd and 3rd cards have a sum of 13.
- The 10th and 11th cards have a sum of 10.
- The 8 is two places to the left of the ace.
- The Jack is two places to the right of the 2.
- The 4 is two places to the right of the 3.
- The 6 is three places to the left of the ace.
- The 9 is somewhere to the right of the 4.
- The 10 is two places to the right of the 5.
- The third card from the right and the third card from the left are both odd.

1st 2nd 3rd 4th 5th 6th 7th 8th 9th 10th 11th 12th 13th

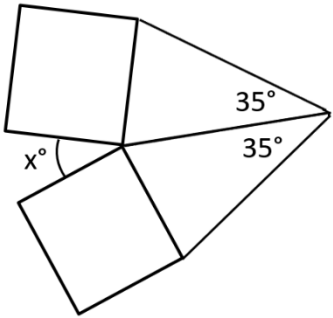
6 8 5 A 10 K 2 Q J 3 7 4 9
1st 2nd 3rd 4th 5th 6th 7th 8th 9th 10th 11th 12th 13th

So, the answer is $10 + 2 + 15 = 27$.

“Math is Cool” Masters -- 2025-26

6th Grade

Triple Jump Solutions

6th	Answer	Solution
1	46 [inches]	Elodie is stacking identical chairs after a meeting. One chair by itself is 31 inches tall. A stack of 4 chairs is 40 inches tall. How tall, in inches, will a stack of 6 chairs be? $31 + 5 \cdot 3 = 46$
2	50 [%]	The 12 faces of a fair 12-sided die are labeled with the numbers 1 through 12. After the die is rolled once, as a percentage, what is the probability that the number showing is a multiple of either 3, 4, or both? Multiples of 3: 3, 6, 9, 12 Multiples of 4: 4, 8, 12 There is 1 that is in both lists, so the probability is $(4 + 3 - 1)/12 = 6/12 = 1/2, = 50\%$
3	36	What is 60% of 120% of 50? 120% of 50 is 60 and 60% of 60 is 36.
4	[x =] 31	The median of the set of integers shown is equal to the mode of the set. What is the value of x? {18, 52, 26, 31, 47, x} Putting the numbers in order 18, 26, 31, x, 47, 52, shows that in order for the median equal the mode, $x = 31$.
5	[x =] 35 [°]	In the diagram, two congruent isosceles triangles share a side, and have one angle measuring 35°. Attached to each triangle is a rectangle that shares a side with the base of the triangle. In degrees, what is the measure of angle x?  The base angles of each isosceles triangle are $(180 - 35)/2 = 72.5$, so $x = 180 - 2(72.5) = 180 - 145 = 35$.

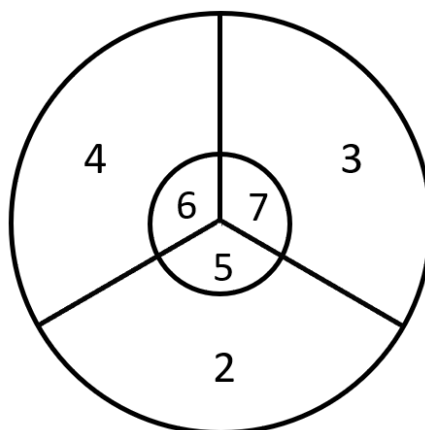
6	32 [cubes]	Forty-six identical cubes are divided into two piles. All of the cubes in the first pile are placed 4 to a bag, and all of the cubes in the second pile are placed 7 to a bag. If a total of 10 bags are used, how many cubes are in the larger of the two piles? Let x be the number of bags with 4 cubes each and y be the number of bags with 7 cubes each. Then $4x + 7y = 46$ and $x + y = 10 \rightarrow 4x + 4y = 40$. Subtracting the two equations results in $3y = 6 \rightarrow y = 2$ and $x = 8$, so the larger pile has $8 \cdot 4 = 32$ cubes in it.																																																	
7	11	In the following puzzle, place the digits 1 through 9 in the outlined empty squares, using each digit exactly once, in order to make the three horizontal and three vertical equations true. Calculations are done from left to right and top to bottom, NOT using the usual "order of operations". For example: $3 + 4 \times 2 = 7 \times 2 = 14$. What is the sum of the two numbers indicated by question marks? The completed table is shown here: <table><tr><td>7</td><td>−</td><td>6</td><td>+</td><td>8</td><td>=</td><td>9</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>−</td><td></td><td></td></tr><tr><td>1</td><td>+</td><td>5</td><td>·</td><td>3</td><td>=</td><td>18</td></tr><tr><td>+</td><td></td><td>+</td><td></td><td>−</td><td></td><td></td></tr><tr><td>9</td><td>+</td><td>4</td><td>·</td><td>2</td><td>=</td><td>26</td></tr><tr><td>=</td><td></td><td>=</td><td></td><td>=</td><td></td><td></td></tr><tr><td>17</td><td></td><td>15</td><td></td><td>3</td><td></td><td></td></tr></table> The sum of the two numbers indicated by question marks is $7 + 4 = 11$.	7	−	6	+	8	=	9	+		+		−			1	+	5	·	3	=	18	+		+		−			9	+	4	·	2	=	26	=		=		=			17		15		3		
7	−	6	+	8	=	9																																													
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9	+	4	·	2	=	26																																													
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17		15		3																																															

8

$$[A + B =]$$

$$1069$$

On the dartboard shown, the outer circle has a radius of 18 cm and the inner circle has a radius of 6 cm. Three radii divide each circle into three congruent regions, with point values shown. The probability that a dart will hit a given region is proportional to the area of the region. When two darts hit this board, the score is the sum of the point values of the regions hit. As a simplified fraction, the probability that the score is odd is A/B . What is $A + B$?



The area of each of the 3 larger regions is $(324\pi - 36\pi)/3 = 96\pi$ and the area of each of the 3 smaller regions is $36\pi/3 = 12\pi$.

The score will be odd when one dart lands in an even region and the other dart lands in an odd region.

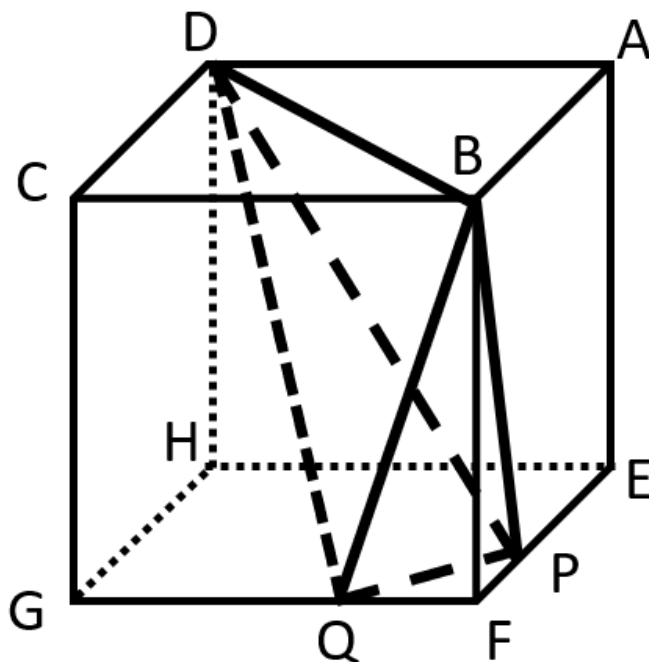
For each throw, $P(\text{even}) = 1/3 + 96/324 = 108/324 + 96/324 = 204/324 = 17/27$ and $P(\text{odd}) = 1 - 17/27 = 10/27$.

$P(\text{EO}) = 17/27 \cdot 10/27 = 170/729$ and $P(\text{OE}) = 10/27 \cdot 17/27 = 170/729$, so $P(\text{score is odd}) = 340/729$, and $340 + 729 = 1069$.

9

3 [cubic
units]

The cube with vertices A, B, C, D, E, F, G, H has edges of length 3 units. Point P is placed so that $FP/FE = 1/3$ and point Q is placed so that $FQ/FG = 1/3$. In cubic units, what is the volume of the tetrahedron with vertices B, D, P, and Q?



To find the volume of the tetrahedron, subtract the volume of the pentagonal pyramid with base HGQPE and vertex D, subtract the volume of the trapezoidal pyramids with base BCGQ and vertex D, and with base AEPB and vertex D, and subtract the triangular pyramid with base FPQ and vertex B.

The volume of the cube is $3^3 = 27$.

The volume of the pentagonal pyramid is **area of the base** $\cdot$ **height of pyramid** $\cdot 1/3 = 8.5 \cdot 3 \cdot 1/3 = 8.5$.

The volume of the 2 trapezoidal pyramids is $2 \cdot$ **area of the base** $\cdot$ **height of pyramid** $\cdot 1/3 = 2 \cdot (1/2)(3 + 2)3 \cdot 3 \cdot 1/3 = 15$.

The volume of the triangular pyramid is **area of the base** $\cdot$ **height of pyramid** $\cdot 1/3 = (1/2)(1 \cdot 1) \cdot 3 \cdot 1/3 = 1/2$.

So, the volume of the tetrahedron is $27 - 8.5 - 15 - 0.5 = 3$.

10	14	Three numbers have a sum of 14 and the sum of their squares is 202. If the product of the three numbers is -42, what is the largest of the three numbers? Let p, q, and r be the three numbers. Then start with the three equations: $p + q + r = 14$ $p^2 + q^2 + r^2 = 202$ $pqr = -42$ $(p + q + r)^2 = 196 \rightarrow p^2 + q^2 + r^2 + 2(pq + qr + pr) = 196 \rightarrow 202 + 2(pq + qr + pr) = 196 \rightarrow -6 = 2(pq + qr + pr) \rightarrow pq + qr + pr = -3$ In a cubic function in the form $ax^3 + bx^2 + cx + d$, when $a = 1$ the sum of the roots is $-b$, the product of the roots is $-d$, and $c = pq + qr + pr$, if p, q, and r, are the roots. So, we can substitute for b, c, and d, to get $x^3 - 14x^2 - 3x + 42 = 0$. Factoring by grouping results in $x^2(x - 14) - 3(x - 14) = 0 \rightarrow (x^2 - 3)(x - 14) = 0 \rightarrow x = \pm\sqrt{3}$ and $x = 14$, so the largest is 14.
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"Math is Cool" Masters -- 2025-26

6th Grade

College Bowl Round #1 Solutions

	Problem	Answer
1	When X is divided by ten and then multiplied by five, the result is twenty. What is X? $20/5 = 4$ and $4 \cdot 10 = 40$	[x =] 40
2	A field has only chickens and sheep. There are seventy-four feet in the field. If there are twenty-one chickens, how many sheep are in the field? $74 - 21 \cdot 2 = 32$ and $32/4 = 8$	8 [sheep]
3	The radius of a circle is thirteen point five centimeters. The circumference of the circle is A times pi centimeters. What is A? $C = 2r\pi = 2 \cdot 13.5 \cdot \pi = 27\pi$, so $A = 27$	[A =] 27
4	What is the smallest three-digit positive integer that is divisible by four, six, and nine? $\text{LCM}(4, 6, 9) = 3^2 \cdot 2^2 = 36$, and the smallest three-digit multiple of 36 is 108.	108
5	A container in the shape of a rectangular prism holds two cubic feet of sand. The area of the base of the container is ninety-six square inches. In feet, what is the height of the container? $2 \text{ cubic feet} = 2 \cdot 1728 = 3456 \text{ cubic inches}$, and $3456/96 = 36$ inches = 3 feet	3 [feet]
6	Two standard six-sided dice are rolled. One is red and one is green. As a simplified fraction, the probability that the number on the green die is less than the number on the red die is A over B. What is A plus B? If R = 6, there are 5 outcomes for green that are less than 6. If R = 5, there are 4 outcomes for green that are less than 5. If R = 4, there are 3 outcomes for green that are less than 4. If R = 3, there are 2 outcomes for green that are less than 3. If R = 2, there is 1 outcome for green that is less than 2. If R = 1, there are 0 outcomes for green that are less than 1. So, $P(G < R) = (1 + 2 + 3 + 4 + 5)/36 = 15/36 = 5/12$, and $5 + 12 = 17$.	[A + B =] 17
7	Celine writes a list of ten consecutive integers. The sum of the least and greatest numbers on her list is ninety-three. What is the smallest number in her list? 42, 43, ..., 51	42

8	The probability that a randomly chosen positive single-digit integer is divisible by two is A over B. What is A plus B? 2, 4, 6, and 8 are all divisible by 2 so the answer is $4/9$, and $4 + 9 = 13$.	$[A + B =] 13$
9	If A star B equals two times A plus three times B, what is the value of four star two? $a * b = 2a + 3b$ $2(4) + 3(2) = 14$	14
10	What is the mean of the numbers twenty-five, eight, seventeen, three, and seven? $(25 + 8 + 17 + 3 + 7)/5 = 12$	12 [= mean]

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6th Grade

College Bowl Round #2 Solutions

	Problem	Answer
1	What is two hundred minus seventy-five plus fifty minus eighty? $200 - 75 + 50 - 80 = 95$	95
2	The median of eight distinct positive integers is ten. What is the smallest possible integer in the set? A possible set could be 1, 2, 3, 9, 11, 12, 13, 14, so the smallest possible number is 1.	1
3	The first three terms of a geometric sequence are two, six, and eighteen. What is the fifth term in the sequence? 2, 6, 18, 54, 162	162 [= 5 th term]
4	How many ways are there to arrange the letters in the word BURRITO, spelled B - U - R - R - I - T - O, if the Rs must not be next to each other? With no restrictions the number of arrangements would be $7!/2! = 2520$. The Rs can be adjacent 6 different ways, and in each of these ways, the remaining 5 letters can be arranged $5! = 120$ ways. So, the final answer is $2520 - 6 \cdot 120 = 1800$.	1800 [ways]
5	A rectangle has its length multiplied by one-point-five and its width multiplied by two-point-five to create a new rectangle. What percent of the area of the original rectangle is the area of the new rectangle? Original area = LW New area = $3L/2 \cdot 5W/2 = 15LW/4 = 3.75LW$ So, New/Original = 3.75 or 375%.	375 [%]
6	Using the digits one through eight exactly once each, form two four-digits numbers with a sum as small as possible. What is the sum? Use the smallest digits for the larger place values: $1458 + 2367 = 3825$	3825 [= sum]
7	A fish's head is six centimeters long. Its tail is the same length as the head plus two-thirds of the body. The body is as long as the head and tail together. In centimeters, how long is the fish? $h = 6, t = h + 2b/3, b = h + t \rightarrow t = 6 + 2b/3, b = 6 + t \rightarrow b = 6 + 6 + 2b/3 \rightarrow b = 12 + 2b/3 \rightarrow 3b = 36 + 2b \rightarrow b = 36, t = 30$, so the length of the fish is $6 + 36 + 30 = 72$.	72 [cm]

8	A rectangular prism has dimensions eleven inches by seventeen inches by nineteen inches. What is the number of cubic inches in the volume of the prism? $11 \cdot 17 \cdot 19 = 187 \cdot 19 = 3740 - 187 = 3553$	3553 [in ³]
9	Athena walks at an average rate of three miles per hour. How many minutes will it take her to walk zero-point-seven-five miles? $3 \text{ mi/h} = 1 \text{ mi}/20 \text{ min} = 0.75 \text{ mi}/15 \text{ min}$	15 [minutes]
10	Solve the equation for X: thirty-two X minus eighty equals one hundred and forty-four $32x - 80 = 144 \rightarrow 32x = 224 \rightarrow x = 7$	[x =] 7

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College Bowl Round #3 Solutions

	Problem	Answer
1	A jar has nine red, four green, and seven purple marbles in it. As a percentage, what is the probability that a randomly chosen marble is not green? $P(\text{not green}) = 16/20 = 4/5, = 80\%$	80 [%]
2	Evaluate X squared minus Y squared if X equals nine and Y equals negative four. $9^2 - (-4)^2 = 81 - 16 = 65$	65
3	How many square inches are in one-point-seven-five square feet? $144 \cdot 1.75 = 252$	252 [in ²]
4	A data set consists of thirteen numbers and has a mean of seventeen. What number, when added to the set, will cause the mean to increase to nineteen? Let N = number of data values and M = the mean. Then $MN \rightarrow (M + 2)(N + 1) = MN + M + 2N + 2$, so $M + 2(N + 1)$ would be the new number, or $17 + 2(13 + 1) = 45$.	45
5	The area of the largest circle that can be inscribed in a regular hexagon is forty-eight pi square centimeters. What is the number of centimeters in the perimeter of the hexagon? The hexagon can be divided into 6 congruent equilateral triangles. If the side length of the hexagon is x, then the height of one of the triangles is $\frac{x\sqrt{3}}{2}$, so the distance from the bottom edge to the top edge of the hexagon, which is also the diameter of the circle, is $x\sqrt{3}$. If the area of the inscribed circle is 48π, then the radius is $4\sqrt{3}$, and the diameter is $8\sqrt{3}$. So, $x\sqrt{3} = 8\sqrt{3} \rightarrow x = 8$, and the perimeter of the hexagon is $8 \cdot 6 = 48$.	48 [cm]
6	The Fibonacci Sequence starts with zero, one, one, two, three, where, starting with term three, each term is the sum of the two previous terms. Let N be the number of terms in the sequence consisting of exactly three digits. What is N? 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, ... Five of these terms are 3-digit integers, so the answer is 5.	[N =] 5

7	What is the greatest common factor of nine hundred sixty, and six hundred thirty? $960 = 2^6 \cdot 3 \cdot 5$ $630 = 2 \cdot 3^2 \cdot 5 \cdot 7$ So, the GCF = $2 \cdot 3 \cdot 5 = 30$	30 [= gcf]
8	Point A with coordinates three comma two and point B with coordinates negative eight comma two are plotted on a coordinate plane. What is the number of units in the distance between point A and point B? $d = \sqrt{(3 - -8)^2 + (2 - 2)^2} = 11$	11 [units]
9	A drawer has two brown socks, four blue socks, three black socks, and six white socks. What is the minimum number of randomly chosen socks needed to ensure that three white socks are selected? In the first 11 socks to be selected, it is possible for there to be 2 brown, 4 blue, 3 black, and 2 white socks. The 12th sock drawn will ensure a minimum of 3 white socks drawn.	12 [socks]
10	What is forty-six times fifty-four? $46 \cdot 54 = (50 - 4)(50 + 4) = 2500 - 16 = 2484$	2484

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6th Grade

College Bowl Round #4 Solutions

	Problem	Answer
1	The area of a triangle is sixty square inches. If the height of the triangle is fifteen inches, what is the number of inches in the base? $60 = b(15)/2 \rightarrow 120 = 15b \rightarrow b = 8$	8 [inches]
2	What is the largest possible number of Wednesdays that can occur during the months of February and March of a given year? There are either $28 + 31 = 59$ or $29 + 31 = 60$ days in February and March combined. If the first day of February is a Wednesday, then the last Wednesday would be $8 \cdot 7 = 56$ days later, so the maximum number of Wednesdays is 9.	9 [Wednesdays]
3	The first three terms of an arithmetic sequence are two, eleven, and twenty. What is the twentieth term in the sequence? $2 + 19(9) = 2 + 171 = 173$	173 [= 20 th term]
4	Biff takes half an hour to trim the hedge. Eho takes twenty-five minutes to trim the same hedge. As a simplified fraction, the fraction of an hour it takes them to trim the hedge if they work together is A over B. What is A plus B? Biff can trim 2 hedges per hour and Eho can trim 2.4 hedges per hour, so together they can trim 4.4 hedges per hour. To trim 1 hedge it would take $1/4.4 = 5/22$ h, and $5 + 22 = 27$.	[A + B =] 27
5	The point with coordinates negative three comma four is on the graph of the line with equation Y equals M X plus twenty-five. What is the value of M? $4 = M(-3) + 25 \rightarrow -21 = -3M \rightarrow M = 7$	[M =] 7
6	In a set of four distinct positive integers the range is twenty and the mean is thirteen. What is the largest possible median for this data set? $4 \cdot 13 = 52$ To maximize the median, make the smallest and largest numbers as small as possible, so 1 and 21. Then the remaining 2 numbers must add up to 30, so the median would be half of 30, or 15.	15
7	It takes nine accountants eight days to file thirteen tax returns. How many accountants would it take to file thirty-nine tax returns in three days? $9A:8D:13T \rightarrow 72A:1D:13T \rightarrow 216A:1D:39T \rightarrow 72A:3D:39T$	72 [accountants]

8	What is the smallest odd composite number with exactly three distinct prime factors? $3 \cdot 5 \cdot 7 = 15$	105
9	If N consecutive numbers are added, the positive difference between the largest and smallest of the consecutive numbers is N minus X. What is X? Let N = 5. An example is 4, 5, 6, 7, 8, and $8 - 4 = 4$ or $N - 1$, so $x = 1$	[x =] 1
10	What is fourteen-thirds divided by two-ninths? $14/3 \div 2/9 = 14/3 \cdot 9/2 = 21$	21

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6th Grade

College Bowl Round #5 Solutions

	Problem	Answer
1	Let A equal twenty-two percent of one hundred and ten. Let B equal fifty-two percent of one hundred and ten. What is B minus A? $A = 0.22 \cdot 110 = 24.2$ $B = 0.52 \cdot 110 = 57.2$ $B - A = 57.2 - 24.2 = 33$	[B - A =] 33
2	How many centimeters are equivalent to nine hundred and fifty millimeters? $950 \text{ mm} = 95 \text{ cm}$	95 [cm]
3	What is six hundred and sixteen divided by eleven? $616/11 = 56$	56
4	The first three terms in an arithmetic sequence are: twenty-one halves, eleven, and twenty-three halves. What is the 6 th term? $21/2, 11, 23/2, 12, 25/2, 13$ Add $\frac{1}{2}$ each time.	13 [= 6 th term]
5	Naveen has quarters and dimes in his pocket in a ratio of Q to D. If instead, the coins were in a ratio of D quarters to Q dimes, the value of his coins would be less by ninety cents. How many more quarters than dimes does he have in his pocket? $25Q + 10D - (10Q + 25D) = 90 \rightarrow 15Q - 15D = 90 \rightarrow Q - D = 6$	6 [more quarters]
6	What is the sum of the factors of eighty-four? $1 + 2 + 3 + 4 + 6 + 7 + 12 + 14 + 21 + 28 + 42 + 84 = 224$	224
7	How many three-digit integers exist, such that their digits form an arithmetic sequence with a positive common difference? Note: the number three hundred twenty-one forms an arithmetic sequence with a negative common difference. $d = 1: 123, 234, 345, 456, 567, 678, 789 - 7$ $d = 2: 135, 246, 357, 468, 579 - 5$ $d = 3: 147, 258, 369 - 3$ $d = 4: 159 - 1$ $1 + 3 + 5 + 7 = 16$	16 [integers]

8	X plus X plus six equals X plus X plus X plus X. What is the value of X plus two? $2x + 6 = 4x$ $2x = 6$ $x = 3$ $x + 2 = 5$	5 [= x + 2]
9	Josh and Grace both draw one card at random from a standard deck. Josh draws first and gets a red card. As a simplified fraction, the probability that Grace also draws a red card is A over B. What is A plus B? $P(\text{Grace gets R}) = 25/51 \text{ and } 25 + 51 = 76$	[A + B =] 76
10	What is the median of the following set of numbers? Fifty, twenty, ninety, forty-four, eighteen, and sixty-two 18, 20, 44, 50, 62, 90 The median is $(44 + 50)/2 = 47$	47 [= median]

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6th Grade

College Bowl Round #6 Solutions

	Problem	Answer
1	The mean of four numbers is twenty-seven. Three of the numbers are fourteen, thirty-five, and fifty-one. What is the fourth number? $4 \cdot 27 = 108$ $108 - 14 - 35 - 51 = 8$	8
2	How many positive two-digit integers have only odd digits? $5 \cdot 5 = 25$	25
3	There are five thousand two hundred and eighty feet in one mile. How many feet are in two point three miles? $5280 \cdot 2 = 10560$ $528 \cdot 3 = 1584$ $10560 + 1584 = 12144$	12144 [feet]
4	The ratio of the areas of two similar rectangles is twelve to seventy-five. As a simplified fraction, the ratio of the rectangles' perimeters is A over B. What is A plus B? (ratio of perimeters) ² = ratio of areas Ratio of perimeters = $\sqrt{\frac{75}{12}} = \sqrt{\frac{25}{4}} = 5/2$, and $5 + 2 = 7$.	[A + B =] 7
5	What is the smallest 3-digit sum of two prime numbers? $97 + 3 = 100$	100
6	A field has only llamas and emus. There are a total of one hundred and thirty-four feet and forty heads in the field. How many more llamas are there in the field than emus? $4l + 2e = 134$ $l + e = 40 \rightarrow 2l + 2e = 80$ $2l = 54 \rightarrow l = 27$ and $e = 13$ $27 - 13 = 14$	14

7	Huey takes twenty-four minutes to read an article. Dewey can read the same article twice as fast as Huey. Louie can read the same article three-fourths as fast as Dewey. In minutes, how long does it take Louie to read the article? Huey takes 24 min For Dewey, twice as fast means half as long, so it takes him 12 min For Louie $\frac{3}{4}$ as fast as Dewey means $\frac{4}{3}$ as long, so it takes him $\frac{4}{3} \cdot 12 = 16$ min	16 [minutes]
8	Six apples cost the same as five pears. Four pears cost the same as eight oranges. How many oranges cost the same as three apples? $6a = 5p$, $4p = 8o$ $24a = 20p$, $20p = 40o$ $24a = 40o$ $3a = 5o$	5 [oranges]
9	The first three terms of a geometric sequence are one hundred sixty-two, one hundred eight, and seventy-two. What is the fifth term in the sequence? 162, 108, 72, 48, 32	32
10	As a simplified fraction, the sum of one-third plus three-fifths plus seven-twelfths is A over B. What is A plus B? $\frac{1}{3} + \frac{3}{5} + \frac{7}{12} = \frac{(20 + 36 + 35)}{60} = \frac{91}{60}$, and $91 + 60 = 151$	[$A + B =$] 151